

ROYAUME DU MAROC • ENSEIGNEMENT SECONDAIRE QUALIFIANT

Pedagogical Lesson Plans

Physics & Chemistry (1st Year Baccalaureate - Sciences)

Experimental Sciences & Mathematical Sciences (A & B)

Prepared by: Professor AYOUB KHAMMOUR

Registration No / N° SOM : 1503811

Approved Textbooks – Pedagogical Orientations – Ministerial Notes 09-142 & 144

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Part 1: Mechanical Work and Energy (34 - 45 Hours)

Targeted Specific Competencies: Solve problems relating to energy conservation and dissipation. Build mechanical setups with safety precautions. Use curve-plotting spreadsheet software and simulation tools. Connect observations to physical principles.

Part Units Breakdown: 1. Rotation around Fixed Axis (7h) • 2. Work and Power of a Force (6h) • 3. Work and Kinetic Energy (4-5h) • 4. Work and Gravitational Potential Energy (5-6h) • 5. Mechanical Energy of a Solid (5-7h) • 6. Work and Internal Energy (6h) • 7. Thermal Energy: Heat Transfer (8h)

Unit 1: Rotational Motion of a Rigid Solid Around a Fixed Axis

7 Hours

Prerequisites:

Motion relativity, average/instantaneous speed, trajectory, moment of a force.

Problem Situation:

Doors and bicycle wheels rotate around fixed axes. How is rotation around a fixed axis characterized, and what links angular speed to linear speed?

Objectives	Contents	Didactic Materials	Activities & Evaluation
- Define rotational motion around a fixed axis. - Define angular position $\theta(t)$ and curvilinear abscissa $s(t) = R \cdot \theta$. - Average and instantaneous angular velocity $\omega = d\theta/dt$ (rad/s). - Relation between linear speed and angular velocity: $v = R \cdot \omega$. - Uniform circular rotation: period $T = 2\pi/\omega$, frequency $f = 1/T$. - Time equation of uniform rotation: $\theta(t) = \omega \cdot t + \theta_0$.	I. Position of Point on Rotating Body (1. Fixed axis rotation, 2. Angular & curvilinear position) II. Angular Velocity (1. Average angular speed, 2. Instantaneous angular speed, 3. Relation $v = R \cdot \omega$) III. Uniform Rotational Motion (1. Definition, 2. Period & frequency, 3. Time equation $\theta(t)$)	- Air cushion table with rotating disc accessory - Spark timer, non-elastic thread, pulleys - Metric circular board, chronometer, Avimeca software	Activities: Record trajectory of points at various radii; verify that $v/R = \omega$ is constant for all points. Write the time equation. Evaluation: Angular velocity calculations, time equation exercises, Exam 1.

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Unit 2: Work and Power of a Force		6 Hours	
Prerequisites: Forces, translation/rotation motion, dot product of vectors.		Problem Situation: We commonly say someone does more work by lifting heavy loads. What is mechanical work in physics, and how is power evaluated?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Work of constant force: $W_{AB}(F) = F \cdot AB \cdot \cos(\alpha)$ (Joules). - Motoring work ($W > 0$), resistive work ($W < 0$), zero work. - Work of weight: $W_{AB}(P) = \pm m \cdot g \cdot h$ (path-independent). - Work of torque in rotation: $W = M_{\Delta} \cdot \Delta\theta$. - Average power $P_m = W / \Delta t$ and instantaneous power $P = F \cdot v$.	I. Work of Forces (1. Constant force in translation, 2. Work of weight, 3. Constant torque in rotation) II. Power of Forces (1. Average power, 2. Instantaneous power in translation $P = F \cdot v$, 3. Instantaneous power in rotation $P = M \cdot \omega$)	- Inclined plane bench, pulleys, dynamometers - Masses, strings, air track with gliders - Winch apparatus for rotational work measurement	Activities: Demonstrate that work of weight depends only on vertical elevation h . Compute power of electric winches. Evaluation: Work calculation problems on inclined planes.

Unit 3: Work and Kinetic Energy		5 Hours	
Prerequisites: Mechanical work, velocity, moment of inertia.		Problem Situation: A blender knife rotates faster and grinds food more rapidly because of acquired kinetic energy. How is kinetic energy related to the sum of work done on a solid?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Kinetic energy in translation: $E_c = 0.5 \cdot m \cdot v^2$ (Joules). - Kinetic energy in rotation: $E_c = 0.5 \cdot J_{\Delta} \cdot \omega^2$. - Moment of inertia J_{Δ} in $\text{kg} \cdot \text{m}^2$. - Kinetic Energy Theorem: $\Delta E_c = E_{c2} - E_{c1} = \Sigma W_1 \rightarrow 2(F_{\text{ext}})$. - Apply theorem to free fall, inclined planes, and rotating solids.	I. Kinetic Energy (1. Solid in translation, 2. Solid in rotation) II. Kinetic Energy Theorem (1. Free fall case, 2. Rectilinear translation case, 3. Rotation case, 4. Statement) III. Applications (1. Friction evaluation, 2. Speed determination)	- Free fall apparatus with digital optical timer - Air track with photogates and hanging load - Flywheel with cord and weights for rotational study	Activities: Verify experimentally that $0.5 \cdot m \cdot v^2 = m \cdot g \cdot h$ in free fall. Measure terminal velocity on inclined plane. Evaluation: Kinetic energy theorem problem solving and Supervised Exam 2.

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Unit 4: Work and Gravitational Potential Energy				5 Hours
Prerequisites: Work of weight, elevation, reference state.		Problem Situation: Water stored behind high dams stores vast energy called gravitational potential energy. How is this energy expressed and utilized?		
Objectives	Contents	Didactic Materials	Activities & Evaluation	
- Understand gravitational potential energy E_{pp}. - Formula: $E_{pp} = m \cdot g \cdot z + C$ (Joules). - Choose a reference state ($E_{pp} = 0$ at $z = 0$). - Fundamental relation: $\Delta E_{pp} = -W_{AB}(P) = m \cdot g \cdot (z_B - z_A)$.	I. Gravitational Potential Energy (1. Definition, 2. Formula $E_{pp} = m \cdot g \cdot z + C$, 3. Reference state) II. Variation of Potential Energy (1. Relation $\Delta E_{pp} = -W(P)$, 2. Applications)	- Textbook, board, computer media - Stands, vertical scale, steel balls	Activities: Establish formula $E_{pp} = m \cdot g \cdot z$ by integration of weight work. Show that choice of reference changes C but not ΔE_{pp} . Evaluation: Potential energy calculations with various reference planes.	

Unit 5: Mechanical Energy of a Solid Body				6 Hours
Prerequisites: Kinetic energy, gravitational potential energy, friction work.		Problem Situation: A pole vaulter converts kinetic energy into potential energy during ascent, and back during descent. Is mechanical energy conserved? What happens when friction occurs?		
Objectives	Contents	Didactic Materials	Activities & Evaluation	
- Define mechanical energy: $E_m = E_c + E_{pp}$ (Joules). - Conservation of mechanical energy: $\Delta E_m = 0$ in absence of friction. - Energy transfer between E_c and E_{pp}. - Non-conservation: $\Delta E_m = W(f) = -Q_{thermal}$.	I. Mechanical Energy (1. Definition $E_m = E_c + E_{pp}$) II. Conservation of E_m (1. Free fall, 2. Frictionless inclined plane, 3. Generalization) III. Non-Conservation of E_m (1. Effect of friction forces, 2. Thermal energy dissipation $\Delta E_m = W(f)$)	- Air cushion table, curved frictionless rail - Pendulum with photogate sensors at various heights - Glider with braking friction pad	Activities: Track oscillating glider; plot $E_c(t)$, $E_{pp}(t)$, and $E_m(t)$; verify E_m is constant without friction and decreases with friction. Evaluation: Mechanical energy conservation problems and Supervised Exam 2.	

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Unit 6: Work and Internal Energy		6 Hours	
Prerequisites: Mechanical work, gas compression, thermal energy.		Problem Situation: Solar radiation heats receiving bodies on Earth. What causes internal temperature rise, and what is the 1st Law of Thermodynamics?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Microscopic interpretation of internal energy U. - Work of pressing force in gas compression: $W = -P \cdot \Delta V$. - Modes of energy transfer: mechanical work W, heat Q, radiation. - First Law of Thermodynamics: $\Delta U = W + Q$.	I. Effects of Mechanical Work (1. Tyndall experiment, 2. Gas compression work $W = -P \cdot \Delta V$) II. Energy Transfer Modes (1. Heat conduction, 2. Convection, 3. Radiation) III. 1st Law of Thermodynamics (1. Internal energy U, 2. Equation $\Delta U = W + Q$)	- Digital thermometers, lead shot in insulated tube - Syringe fitted with temperature and pressure sensors - Polystyrene calorimeter, heating mantle	Activities: Shake lead shot in cylinder; measure temperature rise resulting from mechanical work. Rapidly compress air in syringe and record T increase. Evaluation: Thermodynamic cycle calculations and ΔU problem sets.

Unit 7: Thermal Energy: Heat Transfer		8 Hours	
Prerequisites: Temperature, thermal equilibrium, internal energy.		Problem Situation: Bodies exchange thermal energy until reaching thermal equilibrium. How is heat expressed without phase change, and what is latent heat of melting or boiling?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Heat quantity formula: $Q = m \cdot c \cdot (\theta_f - \theta_i)$ (Joules). - Specific heat capacity c (J/(kg·°C)) & heat capacity C = m·c. - Calorimeter thermal equilibrium: $\sum Q_i = 0$. - Determine calorimeter heat capacity C_{cal} and metal specific heat c. - Latent heat of phase change: $Q = m \cdot L$ (melting, boiling).	I. Heat Transfer without Phase Change (1. Heat quantity Q, 2. Specific heat c, 3. Thermal equilibrium) II. Calorimetry Experimental Study (1. Calorimeter description, 2. Determination of C _{cal} , 3. Measuring metal specific heat) III. Heat Transfer with Phase Change (1. Latent heat L, 2. Melting of ice L _f , 3. Vaporization L _v)	- Adiabatic calorimeter with accessories and stirrer - Digital precision thermometers (0.1°C resolution) - Calibrated metal cylinders (Cu, Al, Fe, Pb), melting ice - Electric immersion heater and precision balance	Activities: Determine calorimeter heat capacity by mixing hot and cold water. Immerse heated copper block to measure c _{Cu} . Melt ice to find L _f . Evaluation: Thermal equilibrium equations and Supervised Exam 3.

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Part 2: Dynamic Electricity & Electromagnetism (23 - 43 Hours)

Competencies: Understand electric and magnetic fields, energy conservation in circuits, Laplace electromagnetic force, electrical safety precautions, and applications in motors and communication.

Part Units Breakdown: 1. Electrostatic Field (4h) • 2. Electrostatic Potential Energy (6h) • 3. Energy Transfer in Circuits (6-7h) • 4. Circuit General Behavior (5-9h) • 5. Magnetic Field (3-4h) • 6. Current Magnetic Field (4-6h) • 7. Laplace's Law (5-7h)

Unit 1: The Electrostatic Field		4 Hours	
Prerequisites: Electrostatic electrification, positive/negative charges, Coulomb interaction.		Problem Situation: Particle accelerators accelerate charged particles using electrostatic fields. What is an electric field, how is it represented, and what is a uniform field?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Coulomb's Law: $F = k \cdot (q_1 \cdot q_2) / d^2$. - Electrostatic field vector E: $F = q \cdot E$ (V/m or N/C). - Superposition principle: $E = E_1 + E_2$. - Field lines and electrostatic field spectra. - Uniform electric field between parallel plates: $E = U / d$.	I. Coulomb Interaction (1. Point charges, 2. Coulomb's Law) II. Electrostatic Field Vector (1. Definition $F = q \cdot E$, 2. Field of point charge, 3. Superposition) III. Field Lines & Uniform Field (1. Field lines mapping, 2. Uniform field $E = U/d$)	- High-voltage generator (HT 0-5kV) - Parallel metal plates capacitor, paraffin oil tank - Semolina grains, electrostatic pendulum	Activities: Visualize field lines using semolina grains aligning in electric field between parallel plates and point charges. Evaluation: Field vector superposition problems and Coulomb calculations.

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Unit 2: Potential Energy of an Electric Charge in Uniform Electrostatic Field				6 Hours
Prerequisites: Electrostatic field, work of constant force, kinetic energy.		Problem Situation: Lightning discharges huge electrostatic energy between clouds and ground. How is electrostatic potential energy defined and related to voltage?		
Objectives	Contents	Didactic Materials	Activities & Evaluation	
- Work of electrostatic force: $W_{AB}(F_e) = q \cdot E \cdot AB = q \cdot (V_A - V_B)$. - Electric potential V (Volts) & equipotential surfaces. - Electrostatic potential energy: $E_{pe} = q \cdot V + C$ (Joules). - Relation $\Delta E_{pe} = -W_{AB}(F_e) = q \cdot (V_B - V_A)$. - Conservation of total energy: $E_T = E_c + E_{pe} = \text{const.}$	I. Work of Electrostatic Force (1. Uniform field work $W = q \cdot E \cdot AB$, 2. Potential difference $V_A - V_B$) II. Electric Potential (1. Equipotential lines, 2. Field-potential relation $E = -dV/dx$) III. Potential & Total Energy (1. Formula $E_{pe} = q \cdot V$, 2. Total energy conservation $\Delta E_T = 0$)	- Conductive paper with electrolytic mapping tray - Voltmeter with exploration probe, DC voltage source - Computer animations of electron deflection	Activities: Map equipotential lines between parallel electrodes with exploration probe. Calculate velocity of accelerated electron ($0.5 \text{ m v}^2 = e \cdot U$). Evaluation: Equipotential mapping exercises and kinetic energy problem sets.	

Unit 3: Energy Transfer in an Electric Circuit				7 Hours
Prerequisites: Current, voltage, Ohm's law, Joulean dissipation.		Problem Situation: Car headlights convert electrical energy into light while losing part as heat. What are energy transfers across receivers and generators, and what is Joule's law?		
Objectives	Contents	Didactic Materials	Activities & Evaluation	
- Electrical energy transferred: $W = U \cdot I \cdot \Delta t$ (Joules). - Electrical power: $P = U \cdot I$ (Watts). - Joule's law in resistors: $W_J = R \cdot I^2 \cdot \Delta t$. - Energy balance of receiver: $W = W_u + W_J = E' \cdot I \cdot \Delta t + r' \cdot I^2 \cdot \Delta t$. - Energy balance of generator: $W_{\text{total}} = W_{\text{supplied}} + W_{\text{internal}}$.	I. Electric Receiver Energy (1. Transferred energy $W = U \cdot I \cdot \Delta t$, 2. Useful energy $W_u = E' \cdot I \cdot \Delta t$, 3. Joule dissipation) II. Joule's Law (1. Joule formula $W = R \cdot I^2 \cdot \Delta t$, 2. Calorimetric verification, 3. Applications & drawbacks) III. Electric Generator Energy (1. Provided energy $U \cdot I \cdot \Delta t$, 2. Total energy $E \cdot I \cdot \Delta t$)	- Calorimeter with heating resistor, thermometer - Small DC motor with pulley and lifting mass - DC power supply, ammeters, voltmeters, rheostat	Activities: Verify Joule's law by heating water in calorimeter: measure electrical energy $W = U \cdot I \cdot t$ vs thermal energy $m \cdot c \cdot \Delta \theta$. Measure motor mechanical efficiency. Evaluation: Energy balance calculations and efficiency determinations.	

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Unit 4: General Behavior of an Electric Circuit		6 Hours	
Prerequisites: Pouillet's law, generator/receiver equations, Joule effect.		Problem Situation: Computer cooling fans dissipate thermal energy produced by circuits. How is electrical energy distributed across a complete circuit, and how is efficiency optimized?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Energy distribution in circuit during duration Δt. - Receiver efficiency: $\eta_r = W_u / W_{rec} = E' / U$. - Generator efficiency: $\eta_g = U / E$. - Overall circuit efficiency: $\eta_{global} = \Sigma W_{useful} / \Sigma W_{total}$. - Factors affecting generator output energy (influence of R_{eq}).	I. Circuit Energy Distribution (1. Receiver energy balance, 2. Generator energy balance, 3. Circuit overall efficiency) II. Influencing Factors (1. Circuit current in resistive load, 2. Influence of EMF and equivalent resistance)	- Batteries with non-negligible internal resistance r - DC motor, resistor decade boxes, multimeters	Activities: Connect motor and variable resistors; calculate power transferred and plot efficiency η vs circuit load. Evaluation: Circuit energy balance calculations and Exam 4.

Unit 5: The Magnetic Field		4 Hours	
Prerequisites: Magnets (North/South poles), compass needle orientation.		Problem Situation: Migratory birds navigate using Earth's geomagnetic field. What is a magnetic field? What are its sources, properties, and how is it measured?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Detect magnetic fields with compass needles. - Magnetic field vector B (Tesla) characteristics. - Measure magnetic field with a Teslameter. - Map field lines and magnetic spectra using iron filings. - Earth's magnetic field components (B_h horizontal, B_v vertical).	I. Magnetic Field Evidence (1. Compass needle detection, 2. Magnet interaction, 3. DC current effect) II. Magnetic Field Vector (1. Vector B definition, 2. Field lines & bar/horseshoe spectra, 3. Superposition) III. Terrestrial Magnetic Field	- Bar magnets, horseshoe magnets, compass needles - Digital Teslameter with Hall probe - Iron filings on glass plates, Earth induction compass	Activities: Sprinkle iron filings on magnet to observe field lines. Measure field intensity B with Teslameter at varying distances. Evaluation: Magnetic vector superposition exercises and Hall probe reading tests.

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Unit 6: Magnetic Field Created by an Electric Current		5 Hours	
Prerequisites: Magnetic field vector, electric current, Teslameter.		Problem Situation: In 1820, Oersted discovered that electric current deflects a compass needle. What are the magnetic fields created by straight wires, coils, and solenoids?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Field of straight conductor: $B = \mu_0 \cdot I / (2\pi \cdot d)$. - Field of flat circular coil: $B = \mu_0 \cdot N \cdot I / (2 \cdot R)$. - Field inside long solenoid: $B = \mu_0 \cdot n \cdot I = \mu_0 \cdot (N/L) \cdot I$. - Right-hand rule and Ampère's observer rule for B direction.	I. Straight Conductor (1. Concentric field lines, 2. Right-hand rule, 3. Formula $B = \mu_0 \cdot I / (2\pi d)$) II. Flat Circular Coil (1. Field lines, 2. Center field formula $B = \mu_0 \cdot N \cdot I / (2R)$) III. Solenoid (1. Uniform interior field, 2. Formula $B = \mu_0 \cdot n \cdot I$)	- Straight vertical wire assembly, flat circular coils - Long study solenoid ($L > 5R$), magnetic needle - High-current DC power supply (0-5A), Teslameter	Activities: Measure B in solenoid vs current I; verify linear relationship $B = k \cdot I$ and calculate permeability μ_0 . Evaluation: Calculations of magnetic fields for various wire geometries.

Unit 7: Electromagnetic Force - Laplace's Law		6 Hours	
Prerequisites: Magnetic field B, current I, vectors.		Problem Situation: Electric motors and loudspeakers convert electrical current into mechanical motion using Laplace electromagnetic forces. What defines the Laplace force?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Laplace force vector: $F = I \cdot (L \times B)$. - Intensity formula: $F = I \cdot L \cdot B \cdot \sin(\alpha)$ (Newtons). - Right-hand rule for Laplace force direction. - Applications: Laplace rails, DC motor, dynamic loudspeaker. - Electromechanical energy conversion.	I. Laplace Force (1. Experimental setup on rails, 2. Laplace's Law statement, 3. Characteristics of F) II. Applications (1. Electrodynamical loudspeaker, 2. DC electric motor) III. Electromechanical Coupling	- Laplace rails apparatus with rolling copper rod - Horseshoe magnet, DC high-current power supply - Dismantled loudspeaker model, demonstration DC motor	Activities: Demonstrate rolling rod displacement when current flows; reverse current or magnetic field to observe motion reversal. Explain voice coil vibration in loudspeaker. Evaluation: Laplace force calculation problems and Supervised Exam 5.

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Part 3: Geometric Optics (14 - 17 Hours)

Competencies: Understand image formation in mirrors and lenses. Apply Snell-Descartes laws. Calculate focal lengths and magnifications. Understand optical instruments like magnifying glasses and microscopes.

Part Units Breakdown: 1. Conditions of Visibility of an Object (4h) • 2. Images Formed by a Plane Mirror (3-4h) • 3. Images Formed by a Thin Converging Lens (7-9h)

Unit 1: Conditions of Visibility of an Object		4 Hours	
Prerequisites: Light propagation, shadow formation, reflection and refraction.		Problem Situation: Descartes stated laws of reflection and refraction in 1637. How does light travel, how does the eye see objects, and how are light rays refracted through plexiglass?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Light ray and light beam concepts. - Conditions for an object to be seen: emitted/diffused light enters the eye. - Snell-Descartes Reflection Law: $r = i$. - Snell-Descartes Refraction Law: $n_1 \cdot \sin(i_1) = n_2 \cdot \sin(i_2)$. - Principle of optical reversibility.	I. Seeing Objects (1. Luminous object, 2. Two visibility conditions) II. The Eye and Light (1. Light ray, 2. Mechanism of vision) III. Reflection and Refraction (1. Reflection law $r = i$, 2. Refraction law $n_1 \sin(i_1) = n_2 \sin(i_2)$, 3. Optical reversibility)	- Laser ray box, semicircular plexiglass block - Circular graduated optical disc (Hartl disc) - Plane mirrors, white screen, blackout curtains	Activities: Direct laser beam at semicircular block; record angles i_1 and i_2 ; verify that $\sin(i_1) / \sin(i_2)$ is constant and equal to refractive index n . Evaluation: Refraction angle calculations and critical angle problems.

Professor: **AYOUB KHAMMOUR** (SOM: 1503811)

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Unit 2: Images Formed by a Plane Mirror				3 Hours	
Prerequisites: Reflection of light, Descartes' laws.		Problem Situation: Rearview mirrors are essential for road safety. Does a driver see everything behind them? What are the characteristics of images in a plane mirror?			
Objectives	Contents	Didactic Materials	Activities & Evaluation		
- Understand plane mirror image: virtual, upright, identical size. - Image is symmetrical to the object relative to mirror plane. - Determine field of view of a plane mirror.	I. Plane Mirror Image (1. Definition, 2. Symmetry & virtual image, 3. Image dimensions) II. Field of View (1. Geometric construction, 2. Applications to rearview mirrors)	- Plane mirror, transparent glass plate, two identical candles - Ruler, pencil, optical sights	Activities: Align two candles on either side of glass plate to prove image position is symmetrical to object. Evaluation: Ray tracing and field of view construction tests.		
Unit 3: Images Formed by a Thin Converging Lens					8 Hours
Prerequisites: Light rays, refraction, plane mirrors.		Problem Situation: A microscope uses two lenses: the objective creates an enlarged inverted image, and the eyepiece magnifies it. How do thin converging lenses form images?			
Objectives	Contents	Didactic Materials	Activities & Evaluation		
- Distinguish converging and diverging lenses. - Optical center O, principal foci F and F', focal length $f' = OF'$. - Optical power / vergence: $C = 1 / f'$ (dioptries δ). - Gauss conditions for sharp stigmatic images. - Descartes conjugation formula: $1/OA' - 1/OA = 1/f'$. - Linear magnification formula: $\gamma = A'B' / AB = OA' / OA$.	I. Thin Lenses (1. Classification, 2. Optical center & foci, 3. Vergence $C = 1/f'$) II. Image Formation (1. Gauss conditions, 2. Three characteristic rays, 3. Real vs virtual images) III. Formulas & Magnification (1. Conjugation formula, 2. Magnification γ , 3. Silbermann & autocollimation methods)	- Graduated optical bench with riders - Converging lenses (+5, +10, +20 dioptries) - Illuminated letter object (F or 1), projection screen, plane mirror	Activities: Project sharp image on screen; measure OA and OA'; verify $1/OA' - 1/OA = 1/f'$. Measure focal length by autocollimation. Evaluation: Ray tracing constructions, magnification calculations, and Exam 6.		

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Semester 1: Chemistry • Measurement in Chemistry (Total: 26 Hours)

Competencies: Master measurement techniques in chemistry. Value environmental protection. Relate physical measurements (conductometry, pressure) to quantities of matter. Carry out direct titrations with precision.

Semester Units Breakdown: 1. Measurement Importance (1h) • 2. Mass, Volume, Pressure, Moles (2h) • 3. Concentration & Solutions (2h) • 4. Tracking Reactions (3h) • 5. Conductometry (7h) • 6. Acid-Base Reactions (3h) • 7. Redox Reactions (4h) • 8. Direct Titrations (4h)

Unit 1: Importance of Measuring Quantities of Matter in Daily Life

1 Hour

Prerequisites:

The mole, concentration, health safety, pollution thresholds.

Problem Situation:

Measuring substances in medicine, nutrition, and environment is crucial. Why measure in chemistry, and what techniques are employed?

Objectives	Contents	Didactic Materials	Activities & Evaluation
- Understand purpose of chemical measurement (informing, monitoring, intervening). - Classify measurement methods: approximate vs precise, continuous vs batch, non-destructive vs destructive. - Interpret chemical warning labels and safety pictograms.	I. Why Measure in Chemistry? (1. Informing the consumer, 2. Quality control & safety, 3. Medical diagnosis & action) II. How to Measure? (1. Approximate vs accurate measurements, 2. Continuous monitoring, 3. Non-destructive methods)	- Mineral water bottles, medication packaging labels - Blood analysis report samples, chemical safety posters - Computer projector for environmental air quality data	Activities: Analyze mineral water label to inventory dissolved ions (Ca^{2+}, Mg^{2+}, Cl^-, SO_4^{2-}). Compare with World Health Organization drinking norms. Evaluation: Label reading and safety pictogram interpretation quiz.

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Unit 2: Mass, Volume, Pressure and Quantity of Matter		2 Hours	
Prerequisites: The mole, molar mass, ideal gas law, pressure.		Problem Situation: Chemists measure mass, volume, and pressure with balances, pipettes, and manometers. How are these quantities converted into moles?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Solids/liquids: $n = m / M = (\rho \cdot V) / M = (d \cdot \rho_{\text{eau}} \cdot V) / M$. - Gases: $n = V / V_m$ and $P \cdot V = n \cdot R \cdot T$ (with $R = 8.314 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$). - Density of gas relative to air: $d = M / 29$. - Boyle-Mariotte's law: $P \cdot V = \text{const}$ at constant T.	I. Quantity of Matter & Mass (1. Solids $n = m/M$, 2. Pure liquids using density) II. Gases & Gas Laws (1. Molar volume V_m , 2. Boyle-Mariotte law, 3. Ideal gas law $P \cdot V = n \cdot R \cdot T$)	- Precision digital balances, graduated flasks - Syringe connected to pressure sensor / digital manometer - Flask with magnesium and sulfuric acid (H_2 gas evolution)	Activities: Verify Boyle-Mariotte law by compressing air in syringe; plot P vs 1/V. Collect H_2 gas and calculate molar volume. Evaluation: Calculations of n from P, V, T and density.

Unit 3: Concentration and Electrolytic Solutions		2 Hours	
Prerequisites: Ionic bonds, polar covalent bonds, dissolution.		Problem Situation: The ocean is a vast ionic electrolyte solution. What happens when an ionic crystal dissolves in water, and how do we calculate actual ion concentrations?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Structure of ionic crystals (e.g. NaCl lattice). - Polarity of water molecule (permanent dipole). - Three stages of dissolution: dissociation, hydration, dispersion. - Dissolution equation: $A_{\alpha} B_{\beta}(s) \rightarrow \alpha A^{+}(\beta+) + \beta B^{-}(\alpha-)$. - Distinguish solute concentration C and actual ion concentration [X].	I. Ionic Solids & Polarity (1. Ionic crystal structure, 2. Polar covalent bond & dipole moment, 3. Water molecule dipole) II. Aqueous Electrolytic Solutions (1. Dissolution mechanism, 2. Hydrated ions) III. Molar Concentrations (1. Solute concentration $C = n/V$, 2. Actual ion concentration $[X_i]$)	- Sodium chloride NaCl, copper sulfate CuSO_4, sugar - Silver nitrate solution AgNO_3, barium chloride BaCl_2 - Conductivity tester with light bulb, beakers, magnetic stirrer	Activities: Test electrical conductivity of distilled water, solid NaCl, and salt water. Calculate $[\text{Fe}^{3+}]$ and $[\text{SO}_4^{2-}]$ in 0.1 mol/L $\text{Fe}_2(\text{SO}_4)_3$ solution. Evaluation: Dissolution equations and ion concentration calculations.

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Unit 4: Tracking the Evolution of a Chemical Transformation		3 Hours	
Prerequisites: The mole, balanced chemical equations, stoichiometry.		Problem Situation: During a reaction, reactants vanish as products accumulate. How is reaction progress x defined, and how do we predict final pressure or gas volume?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Model transformation with balanced equation. - Reaction progress x in moles. - Construct ICE progress table. - Identify limiting reactant and maximum progress x_{max}. - Predict final gas volume V_f and pressure P_f.	I. Chemical Transformation (1. Initial state & final state, 2. Reaction equation) II. Progress of Reaction (1. Definition of x , 2. ICE progress table) III. Final Matter Balance (1. Limiting reactant, 2. Maximum progress x_{max} , 3. Gas evolution)	- Zinc powder, hydrochloric acid solution - Sealed gas generation flask, gas syringe - Precipitation solutions: AgNO_3, NaCl, NaOH, CuSO_4	Activities: React known mass of Zn with excess HCl; measure H_2 gas volume; compare experimental volume with $x_{\text{max}} \cdot V_m$. Evaluation: ICE table construction and limiting reactant exercises.

Unit 5: Determining Quantity of Matter by Physical Measurement: Conductometry		7 Hours	
Prerequisites: Ohm's law, AC current, ionic concentrations, conductance.		Problem Situation: Environmental monitors check river and seawater purity using conductivity meters. What is conductivity, and how is Kohlrausch's law applied?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Conductance $G = 1/R = I/U$ (Siemens S). - Cell constant: $K_{\text{cell}} = S / l$ (m); relation $G = \sigma \cdot K_{\text{cell}}$. - Conductivity σ (S/m) of dilute electrolytic solution. - Kohlrausch's Law: $\sigma = \sum \lambda_i \cdot [X_i]$ (where $[X_i]$ in mol/m^3). - Conductometric calibration curve: $G = f(C)$ or $\sigma = f(C)$.	I. Conductance of Electrolytes (1. Resistance & conductance $G = I/U$, 2. Cell geometry factors, 3. Solution factors) II. Conductivity (1. Definition $\sigma = G \cdot (l/S)$, 2. Molar ionic conductivities λ_i , 3. Kohlrausch's law) III. Calibration Curve (1. Standard series, 2. Finding unknown concentration)	- Digital conductometer with cell (parallel platinum plates) - Standard KCl solutions (0.01, 0.05, 0.1 mol/L) - NaCl solutions of varied concentrations, unknown saline solution	Activities: Measure conductance G of standard NaCl series; plot G vs C calibration line; deduce concentration of unknown saline solution. Evaluation: Kohlrausch calculations (converting mol/L to mol/m^3) and Exam 2.

Level: 1BAC Sciences		Subject: Physics-Chemistry • Chemistry Component	Pedagogical Lesson Plans
Unit 6: Acid-Base Reactions			3 Hours
Prerequisites: pH concept, acidic and basic solutions, ions in water.		Problem Situation: Cyanidin dyes poppy flowers red in acid and cornflowers blue in base. What defines an acid and a base according to Brønsted, and what is an acid-base reaction?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Brønsted-Lowry definition of acid (H^+ donor) and base (H^+ acceptor). - Conjugate acid-base couple HA / A^-. - Proton exchange half-reaction: $HA \rightleftharpoons A^- + H^+$. - Amphoteric species (ampholyte): water H_3O^+/H_2O and H_2O/HO^-. - Balanced acid-base equation: $HA_1 + A_2^- \rightleftharpoons A_1^- + HA_2$. - Colored indicators as acid-base couples (HIn / In^-).	I. Brønsted Acid-Base Theory (1. Acid definition (proton donor), 2. Base definition (proton acceptor)) II. Conjugate Couples (1. Proton half-equation, 2. Ampholytes & water couples, 3. Common couples (CH_3COOH/CH_3COO^-, NH_4^+/NH_3)) III. Acid-Base Reaction & Indicators (1. Proton transfer reaction, 2. Colored indicators)	- Ethanoic acid CH_3COOH, hydrochloric acid HCl, sodium hydroxide $NaOH$ - Ammonia solution NH_3, ammonium chloride NH_4Cl - Bromothymol blue (BBT), methyl orange, phenolphthalein - pH paper, test tubes, dropper pipettes	Activities: Test color of BBT in acidic, neutral, and basic media. Write half-equations and combine for $CH_3COOH + NaOH$ and $NH_4^+ + HO^-$. Evaluation: Writing balanced acid-base reactions and conjugate couples tests.

Level: 1BAC Sciences	Subject: Physics-Chemistry • Chemistry Component	Pedagogical Lesson Plans
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Semester 2: Chemistry • Redox Transformations and Direct Titrations (8 Hours)

Unit 7: Oxidation-Reduction Reactions			4 Hours
Prerequisites: Atoms, ions, valence electrons, metals.		Problem Situation: Iron rusts when exposed to moist air in an oxidation-reduction reaction. What is an oxidant and a reductant? What governs electron transfer in redox reactions?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Oxidant: species that gains electrons (reduction). - Reductant: species that loses electrons (oxidation). - Redox couple Ox / Red and electronic half-equation: $\text{Ox} + n e^- \rightleftharpoons \text{Red}$. - Balance complex redox equations in acidic media (H^+ and H_2O). - Everyday redox applications (batteries, corrosion, bleaching).	I. Redox Concepts (1. Oxidation & reduction, 2. Electron transfer) II. Ox/Red Couples (1. Definition, 2. Electronic half-equation, 3. Common couples (Cu^{2+}/Cu , $\text{Fe}^{3+}/\text{Fe}^{2+}$, $\text{MnO}_4^-/\text{Mn}^{2+}$, I_2/I^-)) III. Balanced Redox Equation	- Copper turnings, zinc granules, iron filings - Copper sulfate solution CuSO_4, silver nitrate AgNO_3 - Acidified potassium permanganate KMnO_4, iron(II) sulfate FeSO_4	Activities: Immerse zinc rod in copper sulfate; observe copper deposition and solution discoloration. Balance reaction between MnO_4^- and Fe^{2+} in acid. Evaluation: Balancing complex redox equations in acidic medium.

Unit 8: Direct Titrations			4 Hours
Prerequisites: Acid-base reactions, redox reactions, stoichiometric ratios, burette.		Problem Situation: To verify water quality or drug purity, chemists perform titrations. What is the equivalence point, and how is unknown concentration calculated?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Principle of titration: rapid, total, unique reaction. - Equivalence definition: reactants mixed in stoichiometric ratio. - Equivalence formula: $n_A / a = n_B / b \Rightarrow C_A \cdot V_A / a = C_B \cdot V_{BE} / b$. - Colorimetric titration (sudden color change at equivalence). - Conductometric titration (slope change of conductivity curve).	I. Principle of Titration (1. Titrated and titrant species, 2. Equivalence state) II. Titration Methods (1. Colorimetric titration, 2. Conductometric titration) III. Equivalence Equation & Precision	- Graduated precision burette (25 mL), volumetric pipette (10 mL) - Magnetic stirrer, Erlenmeyer flasks, conductometer with probe - Solutions: KMnO_4 (known C), Fe^{2+} (unknown C), H_2SO_4, BBT indicator	Activities: Titrate iron(II) with purple KMnO_4 ; observe persistent pink color at equivalence volume V_E ; calculate unknown concentration C_{Fe} . Evaluation: Stoichiometric equivalence calculations and Supervised Exam 4.

Level: 1BAC Sciences	Subject: Physics-Chemistry • Chemistry Component	Pedagogical Lesson Plans
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Semester 2: Chemistry • Organic Chemistry (Total: 15 Hours)

Competencies: Synthesize organic compounds. Apply IUPAC systematic nomenclature. Represent 3D molecular structures. Understand industrial cracking, reforming, and functional groups.

Part Units Breakdown: 1. Expansion of Organic Chemistry (2h) • 2. Organic Molecules & Carbon Skeletons (5h) • 3. Modifying the Carbon Skeleton (3h) • 4. Functional Groups - Alcohols Reactivity (5h)

Unit 1: Expansion and Horizons of Organic Chemistry 2 Hours

Prerequisites:

Covalent bonds, Lewis model, carbon element.

Problem Situation:

Petroleum provides raw materials for millions of organic compounds. What is organic chemistry, and why is carbon unique in building complex chains?

Objectives	Contents	Didactic Materials	Activities & Evaluation
- Historical milestones of organic chemistry (Wöhler synthesis 1828). - Carbon atom tetravalence (4 covalent bonds). - Natural sources: petroleum, natural gas, biomass, photosynthesis. - Importance of synthetic chemistry in medicine, plastics, fibers.	I. Organic Chemistry Overview (1. Definition & carbon role, 2. Natural sources) II. Carbon Atom Properties (1. Tetravalence, 2. Single, double, and triple bonds) III. Societal Impact	- Textbook, molecular model kits, video clips on refining - Samples of plastics (PE, PVC, PS) and synthetic fibers	Activities: Student presentations on history of organic synthesis and petroleum refining fractions. Evaluation: Oral questions on carbon bonding and natural sources.

Unit 2: Organic Molecules and Carbon Skeletons 5 Hours

Prerequisites:

Carbon tetravalence, molecular formulas, structural formulas.

Problem Situation:

Chlorophyll and fuels have complex carbon chains. What are the types of carbon chains, and how are alkanes and alkenes systematically named?

Objectives	Contents	Didactic Materials	Activities & Evaluation
- Saturated vs unsaturated carbon chains. - Straight, branched, and cyclic carbon skeletons. - Skeletal/topological formulas of organic molecules. - Alkanes ($C_n H_{(2n+2)}$) and IUPAC nomenclature rules. - Alkenes ($C_n H_{2n}$) and stereoisomerism (Z / E).	I. Carbon Chains Diversity (1. Saturated & unsaturated, 2. Linear, branched, cyclic, 3. Topological formula) II. Alkanes & Nomenclature (1. General formula, 2. Alkyl groups, 3. Naming branched alkanes) III. Alkenes & Isomerism (1. Double bond $C=C$, 2. (Z) and (E) stereoisomers, 3. Bromine water test)	- Molecular model sets (black carbon, white hydrogen) - Bromine water, cyclohexane, cyclohexene (alkene test) - Test tubes, test tube rack, fume hood	Activities: Build models of butane and 2-methylpropane (structural isomers). Build (Z)-but-2-ene and (E)-but-2-ene. Test alkene unsaturation with bromine water. Evaluation: IUPAC naming worksheets and Z/E configuration tests.

Unit 3: Modifying the Carbon Skeleton			3 Hours
Prerequisites: Alkanes, alkenes, cracking, refining.		Problem Situation: Petroleum fractions undergo chemical transformations to produce high-octane gasoline. How do cracking, reforming, and polymerization modify carbon chains?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Shortening carbon chain: catalytic thermal cracking. - Isomerization, cyclization, and dehydrogenation (reforming). - Lengthening carbon chain: addition polymerization ($n\text{M} \rightarrow [\text{M}]_n$). - Industrial applications: plastics, high-octane automotive fuels.	I. Why Modify Carbon Chains? (1. Producing high-grade fuels, 2. Chemical industry feedstocks) II. Modification Reactions (1. Chain shortening: cracking, 2. Reforming: isomerization & cyclization, 3. Chain lengthening: polymerization)	- Paraffin oil, ceramic catalyst fragments / brick piece - Pyrex test tube, bent delivery tube, bromine water test - Styrene monomer, benzoyl peroxide catalyst	Activities: Perform catalytic cracking of paraffin oil over hot ceramic; collect gas and decolorize bromine water (confirming alkene formation). Polymerize styrene. Evaluation: Equations of cracking and addition polymer repeating unit identification.

Unit 4: Functional Groups - Reactivity of Alcohols			5 Hours
Prerequisites: Alkanes, redox reactions, functional groups.		Problem Situation: Bees communicate with pheromones containing alcohols and aldehydes. How are the three classes of alcohols identified, and how does mild oxidation distinguish them?	
Objectives	Contents	Didactic Materials	Activities & Evaluation
- Identify hydroxyl group -OH and alcohol classes (1°, 2°, 3°). - Mild oxidation of primary alcohol → aldehyde → carboxylic acid. - Mild oxidation of secondary alcohol → ketone. - Tertiary alcohols do not undergo mild oxidation. - Characterization tests: 2,4-DNPH (yellow/orange ppt), Fehling (red ppt), Tollens (silver mirror).	I. Alcohols & Classes (1. General formula R-OH, 2. Primary, secondary, tertiary classes, 3. Nomenclature) II. Mild Oxidation (1. Primary alcohol oxidation, 2. Secondary alcohol oxidation, 3. Tertiary alcohol resistance) III. Identification Tests (1. 2,4-DNPH for carbonyls, 2. Fehling test for aldehydes, 3. Redox equations with MnO_4^-)	- Butan-1-ol (1°), butan-2-ol (2°), 2-methylpropan-2-ol (3°) - Acidified potassium permanganate solution KMnO_4, sulfuric acid 2,4-DNPH reagent, Fehling solution A & B, hot water bath	Activities: Oxidize the 3 alcohols with acidified KMnO_4 . Test products with 2,4-DNPH and Fehling; verify 1° forms aldehyde then acid, 2° forms ketone, 3° does not react. Evaluation: Balancing alcohol oxidation redox equations and end-of-year Supervised Exam 6.